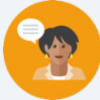


12.5 Pythagorean Theorem and Tangent Lines

Lesson Introduction

 Benchmark	MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.		 Learning Target	I can use the Pythagorean Theorem to solve mathematical and real-world problems involving the length of a tangent segment to a circle.
 Benchmark Coherence	Building On: MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.		Working Toward: N/A	
 Essential Questions	- How can you determine if a line is tangent to a circle? - How can you use the Pythagorean Theorem and tangent lines to calculate line segments?			
 Lesson Narrative	This lesson activates prior knowledge regarding tangent lines to a circle and the Pythagorean Theorem. Students will discover how to utilize the Pythagorean Theorem to determine if a line is tangent to the circle. The primary learning outcome in this lesson focuses on identifying the properties of lines tangent to a circle and using the Pythagorean Theorem and its converse.			
 Component Summary	12.5.1 Warm-Up (~5 min.)	Compose questions about an igloo shelter (<i>SE p. 252</i>)	12.5.3 Guided Practice (10-15 min.)	Using the Pythagorean Theorem and tangent lines (<i>SE p. 254</i>)
	12.5.2 Exploration (15 min.)	Explore Pythagorean Theorem and tangent lines (<i>SE p. 252</i>)	12.5.4 Your Turn! (10 min.)	Individual practice on using the Pythagorean Theorem and tangent lines (<i>SE p. 255</i>)
	12.5.5 Wrap-Up (~5 min.)	Ticket Out the Door demonstrating mastery of learning target (<i>SE p. 256</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Pythagorean Theorem - Tangent - Area - Perimeter		(Optional) Colored Markers or Highlighters	
			Additional Preparation N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may assume all lines tangent to a circle form right angles. - Students may confuse the perimeter and area formulas.	- Students who struggle with the Pythagorean Theorem might need additional support. Consider directing students to a previous lesson, or have a solved Pythagorean Theorem problem available to give them to reference. - Remind students that when finding perimeter and area, they need to include the units with their answers. If the units are not given, instruct them to include the word “units” with their answer.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Clarify, Critique, Connect Consider implementing this routine by allowing students to trade their response in 12.5.3 Guided Practice and 12.5.4 Your Turn! with another student. Students will then ask their partner clarifying questions, as needed. This process will allow students to think more analytically and provide a more detailed mathematical explanation.	MA.K12.MTR.3.1: Mathematical Fluency Mathematical fluency involves being accurate, efficient, and precise while carrying out calculations. By providing students with multiple opportunities to execute procedures and perform calculations, this lesson allows students to accurately determine the length of line segments and determine if a line is tangent to a circle.
 Differentiate Instruction	Read and discuss questions aloud to assist students who benefit from an auditory entry-point. Encourage students to use the diagrams to help make sense of each question. Consider showing students how to color code each expression and tangent line with colored markers or highlighters. To provide a kinesthetic entry-point, consider placing the questions from 12.5.3 Guided Practice and 12.5.4 Your Turn! around the room to create a Gallery Walk.	
Lesson Notes:		

12.5.1 Warm-Up: Ice Problem

Component Overview: Students are shown a picture of an igloo and are asked to write specific questions related to the image.



12.5.1 Warm-Up: Ice Problem

1. An igloo, also known as a snow house or snow hut, is a type of shelter built out of snow.



- a. What is the first question that comes to your mind?
Answers will vary. It is warm? How do they stay warm? Why is the snow house built like a sphere instead of a block?
- b. Write a question that would require someone to use geometry to solve.
Answers will vary. If the circular part of the igloo has a radius of 5 feet (ft), can you find the approximate area of the igloo given it is not a full circle?



This question provides the opportunity for interdisciplinary studies. Consider challenging students by having them incorporate information about igloos to lessons they have studied in their Civics or other Social Studies classes. You could also use the science of how an igloo retains its heat as an extension activity.

12.5.2 Exploration: Pythagorean Theorem and Tangent Lines

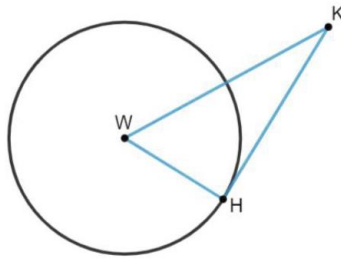
Component Overview: Students explore the relationship between tangent lines of a circle and the Pythagorean Theorem. Students will have the opportunity to collaborate with a partner and other classmates regarding their conclusions.



12.5.2 Exploration: Pythagorean Theorem and Tangent Lines

Work with your partner to complete the following.

- Circle W has point H on the circle. $\overline{WK}$ and tangent $\overline{KH}$ intersect at point K outside the circle.



- Discuss with your partner what can be concluded from the given information. Justify your conclusions. Give at least two conclusions.
Answers will vary. $\overline{WH}$ is a radius because point W is the center of the circle and point H lies on the circle. $\angle KHW$ is a right angle because a tangent is perpendicular to a radius at the point of tangency.



Think-Pair-Share

Consider providing students with the opportunity to use this instructional routine prior to answering the questions. This will allow students to make observations and conclusions in a low-stakes way before sharing with a partner and/or class.

12.5.2 Exploration: Pythagorean Theorem and Tangent Lines (continued)

- b. Ask a classmate for two conclusions they drew. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conclusion	My Summary of Their Reason	Initials
<i>Answers will vary.</i>		

- c. Discuss with your partner what information is needed to determine the length of $\overline{WK}$.
The length of $\overline{HW}$ and $\overline{KH}$.
- d. Explain why any triangle drawn, where the radius and a tangent are the legs, will result in a right triangle.
If a line is tangent to a circle and perpendicular to the radius at the point of tangency, it will form a right angle.

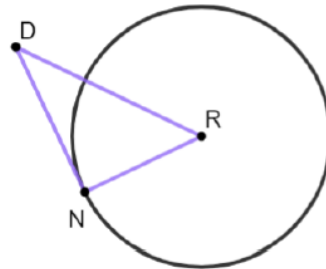


Consider prompting students with questions before answering question 1d:

- What is the relationship between a tangent line and the radius of a circle?
- Will this always be true? Why or why not?
- Can you create an example where the radius and a tangent line will not result in a right triangle?

12.5.2 Exploration: Pythagorean Theorem and Tangent Lines (continued)

2. Circle R is shown with point D outside the circle, where point N is on the circle. $DN = 10$, $NR = 8$, and $DR = 4\sqrt{10}$.



Students may believe that the given line segment is automatically tangent, therefore creating a right angle and right triangle. During question 2, remind students that in Geometry, not all diagrams are drawn to scale and that we cannot assume something is true without completing the math to prove it is true.

- a. Use the Pythagorean Theorem to determine if $\overline{DN}$ is a tangent to circle R .

$$10^2 + 8^2 = 2\sqrt{41}, \quad 2\sqrt{41} \neq 4\sqrt{10}$$

- b. Complete the statements.

$\triangle DRN$ ☐ is
☒ is not a right triangle. Therefore, $\overline{DN}$

☐ is
☒ is not a tangent line to circle R .

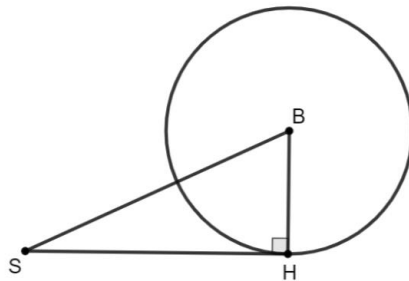
12.5.3 Guided Practice: Pythagorean Theorem and Tangent Lines

Component Overview: Students will use their knowledge of the Pythagorean Theorem and tangent lines to prove a line is tangent to a circle, and to find lengths of line segments. Students will use the lengths of line segments found to calculate the areas and perimeters of triangles.



12.5.3 Guided Practice: Pythagorean Theorem and Tangent Lines

1. Circle B has point H on the circle and point S outside the circle.
 $m\angle SHB = 90^\circ$,
 $SB = 13$, and
 $BH = 5$.



- a. What is the length of $\overline{SH}$?

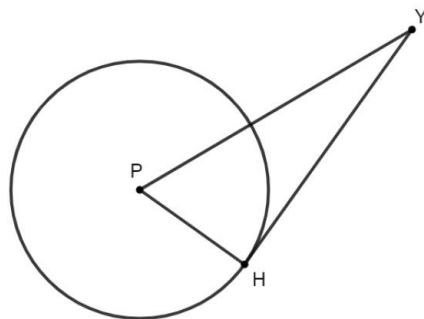
$$SH^2 + 5^2 = 13^2, \quad SH = 12$$

- b. What is the area of $\triangle SBH$?

$$\frac{1}{2}(5)(12) = 30 \text{ units squared}$$

2. Circle P has point H on the circle and point Y outside the circle, where $\overline{PY}$ and tangent $\overline{HY}$ intersect. $PH = 10$, and $HY = 17$.

What is the length of $\overline{PY}$?



$$10^2 + 17^2 = PY^2, \quad PY = \sqrt{389}$$



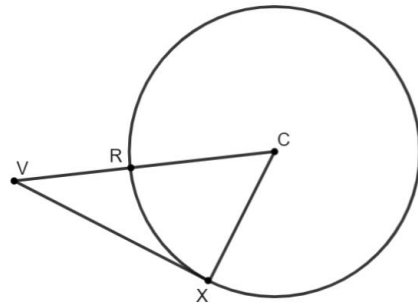
Use inquiry to guide students through question 1 before they try questions 2 and 3 on their own:

- How can you determine that a line is tangent to a circle?
- What specific information do you need for this?
- What is the difference between a tangent line and a secant line?
- How can tangent lines of a circle help you to find the area and perimeter of a triangle?

12.5.3 Guided Practice Pythagorean Theorem and Tangent Lines (continued)

3. Circle C has points R and X on the circle and point V outside the circle, where $\overline{VC}$ and tangent $\overline{VX}$ intersect at point V . $VX = 24$ and $VR = 16$.

- a. What is the length of $\overline{CX}$?



Let x equal the length of the radius, $\overline{CR}$ and $\overline{CX}$.

$$(16 + x)^2 - 24^2 = x^2, \quad x = 10$$

$$CX = 10$$

- b. What is the length of $\overline{VC}$?

$$VC = 26$$

- c. What is the perimeter of $\triangle VCX$?

$$\text{Perimeter} = 60 \text{ units}$$



Provide the question stems that were used in question 1 for students to use on their own as additional scaffolding for question 3. Alternatively, prompt students to use the Study Expert video for hints while working on question 3.

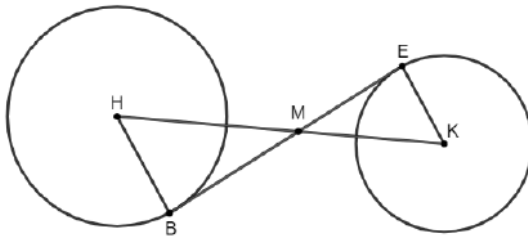
12.5.4 Your Turn!

Component Overview: Students work independently to use tangent lines to find the perimeter and area of triangles, and to identify an error in work.



12.5.4 Your Turn!

- Circle H has point B on the circle and circle K has point E on the circle. $\overline{HK}$ and tangent $\overline{EB}$ intersect outside the circles at point M . $m\angle HBM = m\angle MEK = 90^\circ$, $HB = 6$, $BM = 8$, and $MK = 8$.



- Explain why $\triangle HMB \sim \triangle KME$.

Since $m\angle HBM = m\angle MEK$, they are congruent to each other. $\angle HMB \cong \angle KME$ by the vertical angles theorem. Therefore, the two triangles are similar by AA similarity.

- What is the length of $\overline{HK}$?

$$6^2 + 8^2 = HM^2, \quad HM = 10$$

$$HM + MK = HK, \quad HK = 18$$

- What is the perimeter of $\triangle KME$?

$$\frac{HM}{MK} = \frac{HB}{EK}, \quad EK = 4.8$$

$$\frac{HM}{MK} = \frac{BM}{EM}, \quad ME = 6.4$$

$$\text{Perimeter of } \triangle KME = 19.2 \text{ units}$$

- What is the area of $\triangle KME$?

$$\frac{1}{2}(EK)(ME) = 15.36 \text{ square units}$$



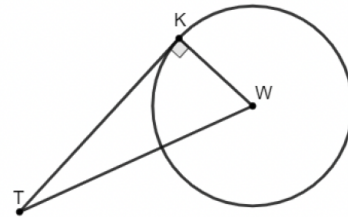
Students may struggle with how to find the perimeter in question 1. Consider reminding students how they can use proportions to determine if triangles are similar, and then how to apply this information to solve question 1c.



Students who see this image as a bicycle may benefit from visualizing the dimensions as a bicycle to find the perimeter.

12.5.4 Your Turn! (continued)

2. Grace and Zackary are finding the length of tangent $\overline{TK}$ of circle W , where K is a point on the circle. $\overline{TW}$ and tangent $\overline{TK}$ intersect at point T outside the circle. $TW = 18$, and $KW = 7$.



Grace's and Zackary's work is shown.

Grace's Work	Zackary's Work
$7^2 + x^2 = 18^2$	$7^2 + 18^2 = x^2$
$49 + x^2 = 324$	$49 + 324 = x^2$
$x^2 = 275$	$x^2 = 373$
$x = 5\sqrt{11}$	$x = \sqrt{373}$

Write a note to the person whose work is incorrect explaining their mistake.

**Means
I
Start
To
Acquire
Knowledge
Experience
Skills**

Answers will vary. Zackary, you were on the right track to recognize that you can use the Pythagorean Theorem to find the length of tangent $\overline{TK}$. However, $\overline{TW}$ is the hypotenuse of the right triangle, not a leg. Next time, it may help to label the triangle first to ensure you substitute the values in for the correct variables.



MA.K12.MTR.4.1: Engage in discussions that reflect on the mathematical thinking of self and others. Provide opportunity for students to get creative with their response. Some students might prefer to write a note, while other students might better engage or activate agency by making a video using technology, or social media.

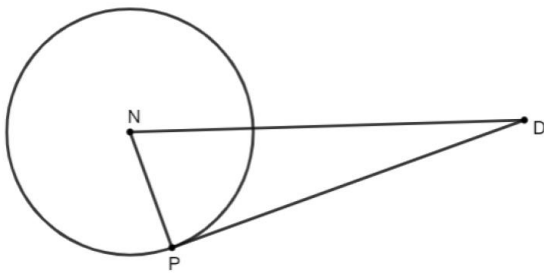
12.5.5 Wrap-Up: Ticket Out the Door

Component Overview: Students find the length of a line segment to reflect their level of understanding of the content of this lesson.



12.5.5 Wrap-Up: Ticket Out the Door

- Circle N has point P on the circle and point D outside the circle at the intersection of $\overline{DN}$ and tangent $\overline{DP}$.
 $NP = 2$ and $DP = 6$.



What is the length of $\overline{ND}$?

$$2^2 + 6^2 = ND, \quad ND = 2\sqrt{10}$$



Consider identifying specific strands from 12.5.5 Wrap-Up to inform small groups in Lesson 6:

- Identifying key measurements
- Labeling the diagram
- Explanation of how you found $\overline{ND}$